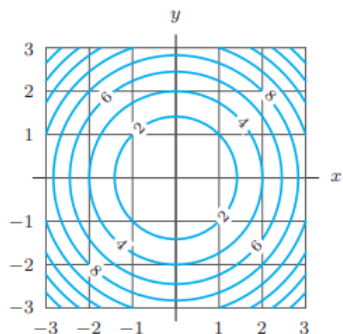


Math 2551 Worksheet: Gradient and Directional Derivatives

1. Use the contour diagram of the differentiable function f given below to decide if the specified directional derivative is positive, negative, or approximately zero.



- At the point $(-2, 2)$ in the direction $\mathbf{i}$
 - At the point $(0, -2)$ in the direction $\mathbf{j}$
 - At the point $(-1, 1)$ in the direction $\mathbf{i} + \mathbf{j}$
 - At the point $(-1, 1)$ in the direction $-\mathbf{i} + \mathbf{j}$
 - At the point $(0, -2)$ in the direction $\mathbf{i} - 2\mathbf{j}$
- Let $f(x, y) = xy$. Sketch the curve $f(x, y) = -4$ together with $\nabla f(2, -2)$ and the tangent line at $(2, -2)$. Then, find an equation for the tangent line. What do you notice?
 - Find the derivative of $g(x, y) = \frac{x - y}{xy + 2}$ at $(1, -1)$ in the direction of $\langle 12, 5 \rangle$.
 - Suppose you are climbing a hill whose shape is given by the equation

$$z = 1000 - 0.005x^2 - 0.01y^2,$$

where x , y , and z are measured in meters, and you are standing at a point with coordinates $(60, 40, 966)$. The positive x -axis points east and the positive y -axis points north.

- If you walk due south, will you start to ascend or descend? At what rate?
 - If you walk northwest, will you start to ascend or descend? At what rate?
 - In which direction is the slope largest? What is the rate of ascent in that direction?
- Let $f(x, y) = -x^2y + xy^2 + xy$ and $P = (2, 1)$.
 - Find the direction of maximal increase of f at P .
 - What is the maximum rate of change of f at P ?
 - Find the direction of maximal decrease of f at P .
 - Find a direction $\mathbf{u}$ such that $D_{\mathbf{u}}f(P) = 0$ (note this forces $\mathbf{u}$ to be a unit vector!).